FitCheck — Logical Relational Database Schema Diagram (Physical Design)

Normalized Relational Model (3NF) | Foreign Key Constraints, Integrity Rules, Unique Keys, and Weak Entity Mappings

1 : N (CASCADE)

users (Strong Entity Table)

PK user_id	INT AUTO_INCREMENT
name	VARCHAR(100) NOT NULL
email [UQ]	VARCHAR(191) UNIQUE
password_hash	VARCHAR(255) NOT NULL
role	ENUM('user','admin')
is_active	BOOLEAN DEFAULT TRUE
created_at	TIMESTAMP DEFAULT NOW

1 : N

1 : 1 (CASCADE)

body_profiles (Weak Entity Mapping)

PK profile_id	INT AUTO_INCREMENT
FK user_id [UQ] -> users	INT (CASCADE)
height	DECIMAL(5,2) NULL
waist	DECIMAL(5,2) NULL
hip	DECIMAL(5,2) NULL
thigh	DECIMAL(5,2) NULL
inseam	DECIMAL(5,2) NULL
preferred_fit	VARCHAR(20) DEFAULT 'Regular'
updated_at	TIMESTAMP DEFAULT NOW

purchases (Associative Entity)

PK purchase_id	INT AUTO_INCREMENT
FK user_id -> users	INT (CASCADE)
FK garment_size_id -> sizes	INT (RESTRICT)
purchase_date	TIMESTAMP DEFAULT NOW
outcome	ENUM('kept','returned')
fit_satisfaction	TINYINT (1-5)
fit_preference	VARCHAR(20)
notes	TEXT NULL

fit_reviews (Associative Review Entity)

PK review_id	INT AUTO_INCREMENT
FK user_id -> users	INT (CASCADE)
FK garment_id -> garments	INT (RESTRICT)
FK size_id -> garment_sizes	INT (RESTRICT)
overall_fit	ENUM('True to Size',...)
waist_fit, hip_fit, length_fit	VARCHAR(20)
review_text	TEXT NOT NULL
rating	TINYINT (1-5)
created_at	TIMESTAMP DEFAULT NOW

1 : N (RESTRICT)

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brands (Strong Entity Table)

PK brand_id	INT AUTO_INCREMENT
brand_name [UQ]	VARCHAR(100) UNIQUE
sizing_system	VARCHAR(50) DEFAULT 'US'
return_rate (Virtual)	DECIMAL(4,2) COMPUTED

1 : N

garments (Strong Entity Table)

PK garment_id	INT AUTO_INCREMENT
FK brand_id -> brands	INT (CASCADE)
garment_name	VARCHAR(150) NOT NULL
garment_category	ENUM('Jeans','Shirt',...)
fit_type	VARCHAR(50)
description	TEXT NULL

1 : N (IDENTIFYING CASCADE)

garment_sizes (Weak Entity Mapping)

PK garment_size_id	INT AUTO_INCREMENT
FK garment_id -> garments	INT (CASCADE)
DISCRIM size_label [PARTIAL KEY]	VARCHAR(20)
UQ(garment_id, size_label)	COMPOSITE CANDIDATE PK
is_available	BOOLEAN DEFAULT TRUE
stock_quantity	INT DEFAULT 0

1 : N (IDENTIFYING CASCADE)

garment_measurements (Weak Entity)

PK measurement_id	INT AUTO_INCREMENT
FK garment_size_id -> sizes	INT (CASCADE)
DISCRIM measurement_type	VARCHAR(30)
UQ(garment_size_id, measurement_type)	COMPOSITE KEY
measurement_value	DECIMAL(5,2) NOT NULL
unit	VARCHAR(10) DEFAULT 'in'

Relational Schema Architectural Legend & Course Mapping Principles:

- **Solid Border Table:** Autonomous Base Table (Strong Entity with Independent Primary Key e.g., users, brands, garments)
- **Dashed Border Table:** Weak Entity Relational Mapping (Existence-dependent table identified by Owner FK + Discriminator)
- PK **Primary Key (PK):** Surrogate primary key guaranteeing physical referential efficiency (AUTO_INCREMENT)
- FK **Foreign Key (FK):** Referential integrity link to referenced parent primary key (with CASCADE or RESTRICT rules)
- DISCRIM **Discriminator:** Weak entity partial key (e.g. size_label, measurement_type) forming composite candidate key with Owner FK

Weak Entity Academic Resolution & Relational Correspondence:

- GARMENT_SIZE cannot exist without an owning GARMENT; a size label like "32W 30L" has no standalone meaning.
- Conceptual Key: Composite {garment_id, size_label}. Relational implementation preserves this via UQ(garment_id, size_label).
- Surrogate garment_size_id is retained to prevent bulky compound foreign keys in child tables (purchases, measurements).
- GARMENT_MEASUREMENT is a second-order weak entity dependent on GARMENT_SIZE via UQ(garment_size_id, measurement_type).
- Foreign Key cascades ensure full existence dependency: Deleting a GARMENT deletes all its sizes and all their measurements.